

BLM 230 Bilgisayar mimarisi - ÖDEV-2

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P1) $16 \text{ KB} = 2^4 \cdot 2^{10} = 2^{14}$ önbellek boyutu byte

1 satır 16 byte
? 2^{14} byte

$\frac{2^{14}}{2^4} = ? \cdot 2^{10}$ satır/line

$128 \text{ MB} = 2^7 \cdot 2^{20} = 2^{27}$ byte

27 bit adres

set sayısı = $\frac{\text{line sayısı}}{\text{way}} \Rightarrow \frac{2^{10}}{2^2} = 2^8$ 8 bit set

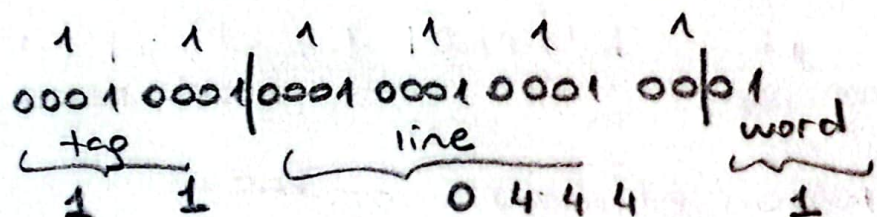
adres = [tag + set + word]
s+w [s-d, d, w]
27 = 15, 8, 4

16 byte = 2^4 u=w
 $\log_2(16)$ block size

P2) a) Direct mapped cache

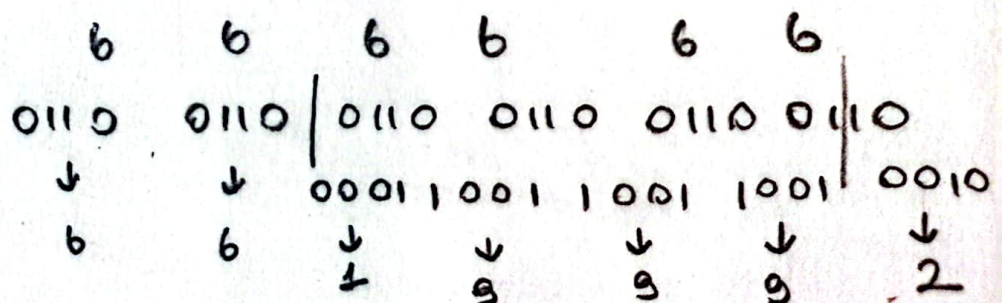
adres = [Tag (8 bit) | line (14 bit) | word (2 bit)]

1 Hex 4 bit
6 Hex 24 bit



111111 → 110441

666666 → 6619992



BBBBBB

1011	1011	1011	1011	1011	1011
B	B	0010	1110	1110	1110
		2	E	E	E
					3

BBBBBB → 2EEF3

EEEE99

1110	1110	1110	1001	1001	1001
E	E	0011	1010	0110	0110
		3	A	6	6
					1

EEEE99 → 3A661

b) associative cach adres = [tag(22), word(2)]

111111	→	0001	0001	0001	0001	0001	0001
		0000	0100	0100	0100	0100	0100
		0	4	4	4	4	4
							1

666666	→	0001	1001	1001	1001	1001	1001
		1	9	9	9	9	9
							2

BBBBBB	→	0010	1011	1011	1011	1011	1011
		2	E	E	E	E	E
							3

EEEE99	→	1110	1110	1110	1001	1001	1001
		0011	1011	1011	1010	0110	0110
		3	B	B	A	6	6
							1

Tag Hex

word Hex

c) two way set Associative

adres = [tag(967) | set(1367) | word(267)]

tag	line/set	word
0100	0100	100
2	2	4
0100	0100	0100
0	4	4
0100	0100	0100
4	4	4
0100	0100	0100
4	4	4
0100	0100	0100
4	4	4

111111 → 22404441

tag	line/set	word
01100	1100	1100
0	0	0
1100	1100	1100
1	9	9
1001	1001	1001
9	9	9
1001	1001	1001
9	9	9
1001	1001	1001
9	9	9
1001	1001	1001
9	9	9

666666 → 0CC19992

Tag	line/set	word
000	1011	1011
1	7	7
000	0	1110
0	E	1110
000	0	1110
0	E	1110
000	0	1110
0	E	1110
000	0	1110
0	E	1110

BBBBBB → 1770EE3

tag	line/set	word
1110	1110	1110
1	D	D
1110	1110	1110
1	A	6
1001	1001	1001
6	6	6
1001	1001	1001
6	6	6
1001	1001	1001
6	6	6

EEEE99 → 1DD1A661

P3) $16K = 2^{10} \cdot 2^4 = 2^{14}$ byte cache
32 bit mikroizlemci

Satır boyutu : 2 quadword 9AC755 nereye ?

$$\text{adres} = \begin{bmatrix} \text{tag} & \text{set} & \text{word} \\ 32 & s-d & d & 4 \end{bmatrix}$$

1 quadword = 8 byte

line size : 16 byte = 2^4 w=4

$$\frac{\text{cache byte}}{\text{line byte}} = \frac{2^{14}}{2^4} = 2^{10} \text{ satır}$$

$$\frac{2^{10}}{4 \text{ way}} = 2^8 \text{ 8 bit (set)}$$

$$\text{tag, set, word} \quad 32 = 20 + 8 + 4$$

Hex: 9AC7558E

Binery: 1001 1010 1100 0111 0101 | 0101 1000 | 1110

$$\text{SET: } 01011000 = (2^3) + (2^4) + (2^6) = 8 + 16 + 64 = 88.$$

9AC7558E adresindeki kelime, ön bellekte

88. kümeye SET 88'e eşleşir. Bu küme içinde herhangi bir yere yerleştirilir.

P4) $s+w = s-r, r, w$ direct mapping
 $16 = s-r, r, 3$
bit

a)

$$\text{line/blok size} = 8 = 2^3 \quad w = 3 \text{ bit}$$

$$\text{Satır sayısı} = 32 \quad 2^5 \quad r = 5 \text{ bit}$$

tag	line	word
8	5	3

b) Adresler hangi satırda line bitinde saklanır.

0001 0001 00011 011 → 3. satır

11000011 00110 100 → 6. satır

11010000 00011 101 → 3. satır

10101010 10101 010 → 21. satır

c) Bir blok ön belleğe alındığında bloğun tamamı gelir.

Bloktaki adresler

0001	1010	0001	1000	offset 0
0001	1010	0001	1001	offset 1
0001	1010	0001	1010	⋮
0001	1010	0001	1011	⋮
0001	1010	0001	1100	⋮
0001	1010	0001	1101	⋮
0001	1010	0001	1110	⋮
0001	1010	0001	1111	offset 7

d) Ön bellek kapasitesi = $\frac{\text{satır}}{\text{sayısı}} \times \frac{\text{satır}}{\text{boyutu}}$

$$= 32 \times 8$$

$$= 256 \text{ bayt.}$$